

Disproof of Conjecture 3486: a separating pair on seven vertices

gaochengzhecpu (AI-assisted submission)

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Claim being disproved. The conjecture asserts that the smallest pair of nonisomorphic four-critical graphs with identical single-vertex-deletion chromatic spectra has thirteen vertices. We exhibit such a pair on seven vertices. Here *four-critical* has its usual strong meaning: the graph has chromatic number four and every proper subgraph is three-colorable.

The graphs. Both graphs have vertex set $\{0, \dots, 6\}$. Let

$$E(G) = \{02, 03, 05, 06, 12, 13, 14, 23, 45, 46, 56\},$$

$$E(H) = \{01, 02, 04, 05, 12, 13, 15, 23, 24, 36, 46, 56\}.$$

The first graph is a Hajós join of two copies of K_4 : delete edges 01 and 04 in the copies on $\{0, 1, 2, 3\}$ and $\{0, 4, 5, 6\}$, and insert edge 14. The second is the Mycielski graph of K_3 : vertices 0, 1, 2 form the original triangle, vertices 3, 4, 5 copy their respective neighborhoods, and vertex 6 is adjacent to 3, 4, 5.

Chromatic number four. In any three-coloring of the subgraph of G on $\{0, 1, 2, 3\}$, vertices 2, 3 have different colors and both 0, 1 are adjacent to both of them. Thus 0, 1 must have the same third color. The same argument on $\{0, 4, 5, 6\}$ forces 0, 4 to have the same color. This contradicts edge 14. For H , the original triangle uses all three colors. Vertex 3 must then have the color of 0, vertex 4 the color of 1, and vertex 5 the color of 2. Vertex 6 sees all three colors, again a contradiction. In contrast, the following vectors, in vertex order 0, $\dots$, 6, are proper four-colorings:

$$c_G = (0, 0, 1, 2, 1, 2, 3), \quad c_H = (0, 1, 2, 0, 1, 2, 3).$$

Hence $\chi(G) = \chi(H) = 4$.

Certificates for every single deletion. Each vector below uses only colors 0, 1, 2. In the vertex-deletion table the entry at the deleted vertex is ignored (denoted by *). The listed edge sets make every certificate directly checkable.

Deleted vertex	coloring of $G - x$	coloring of $H - x$
0	(*, 0, 1, 2, 1, 0, 2)	(*, 0, 1, 2, 0, 2, 1)
1	(0, *, 1, 2, 0, 1, 2)	(0, *, 1, 0, 2, 2, 1)
2	(0, 1, *, 2, 0, 1, 2)	(0, 1, *, 0, 2, 2, 1)
3	(0, 1, 2, *, 0, 1, 2)	(0, 1, 2, *, 1, 2, 0)
4	(0, 0, 1, 2, *, 1, 2)	(0, 1, 2, 0, *, 2, 1)
5	(0, 0, 1, 2, 1, *, 2)	(0, 1, 2, 0, 1, *, 2)
6	(0, 0, 1, 2, 1, 2, *)	(0, 1, 2, 0, 1, 2, *)

Deleted edge of G	coloring of $G - e$	Deleted edge of H	coloring of $H - e$
02	(0, 1, 0, 2, 0, 1, 2)	01	(0, 0, 1, 2, 2, 1, 0)
03	(0, 1, 2, 0, 0, 1, 2)	02	(0, 1, 0, 2, 1, 2, 0)
05	(0, 0, 1, 2, 1, 0, 2)	04	(0, 1, 2, 0, 0, 2, 1)
06	(0, 0, 1, 2, 1, 2, 0)	05	(0, 1, 2, 0, 1, 0, 2)
12	(0, 1, 1, 2, 0, 1, 2)	12	(0, 1, 1, 0, 2, 2, 1)
13	(0, 1, 2, 1, 0, 1, 2)	13	(0, 1, 2, 1, 1, 2, 0)
14	(0, 0, 1, 2, 0, 1, 2)	15	(0, 1, 2, 0, 1, 1, 2)
23	(0, 1, 2, 2, 0, 1, 2)	23	(0, 1, 2, 2, 1, 2, 0)
45	(0, 0, 1, 2, 1, 1, 2)	24	(0, 1, 2, 0, 2, 2, 1)
46	(0, 0, 1, 2, 1, 2, 1)	36	(0, 1, 2, 0, 1, 2, 0)
56	(0, 0, 1, 2, 1, 2, 2)	46	(0, 1, 2, 0, 1, 2, 1)
		56	(0, 1, 2, 0, 1, 2, 2)

Every proper subgraph omits either a vertex or, if it has every vertex, an edge. Restrict the corresponding coloring from these tables to that subgraph. This proves that every proper subgraph of either graph is three-colorable. Both graphs are therefore four-critical in the strong sense.

Identical vertex-deletion spectra. The graph G contains the two vertex-disjoint triangles 123 and 456, so at least one survives every single vertex deletion. For H , after deleting 0, 1, 2 respectively, the triangles 123, 024, 015 survive; after deleting any of 3, 4, 5, 6, the triangle 012 survives. Thus all these vertex-deleted graphs require at least three colors. Combined with the table, this gives

$$(\chi(G - x))_{x=0}^6 = (3, 3, 3, 3, 3, 3, 3) = (\chi(H - x))_{x=0}^6.$$

Nonisomorphism and contradiction. There are eleven edges in G and twelve in H , so the graphs are nonisomorphic. An additional structural distinction, used in Lean, is that G has two vertex-disjoint triangles and H does not. Indeed, vertex 6 belongs to no triangle and $\{3, 4, 5\}$ is independent. Every triangle of H therefore uses at least two of the three vertices $\{0, 1, 2\}$, so any two such triangles intersect. We have exhibited a valid pair at order $7 < 13$, disproving the claimed minimum thirteen. We do not claim to determine the true minimum order.

Lean verification and semantic correspondence. The self-contained file `lean/Main.lean` imports only `Std`. It defines the actual graphs by the displayed edge lists. The definitions `Proper`, `ProperDeleted`, and `ProperEdgeDeleted` express proper colorings before deletion and after deleting an arbitrary vertex or edge. All color vectors in the tables are verified for every edge; exhaustive kernel-checked tests also exclude every three-coloring of the original graphs and every two-coloring of a vertex deletion. The theorem `tuple_eta` proves that the finite tuple enumeration covers every coloring function, so these tests exclude arbitrary colorings, not only a selected sample.

The theorem `proper_subgraph_bridge` formally restricts the single-deletion colorings to every proper subgraph. Consequently `G_four_critical` and `H_four_critical` establish strong four-criticality. The theorem `same_deletion_spectrum` proves the exact chromatic number three, including minimality, for every vertex deletion of both graphs. The theorem `graphs_not_isomorphic` excludes an actual adjacency-preserving vertex bijection by the disjoint-triangle invariant. Finally, `not_minimum_thirteen_consequence` refutes the necessary order-seven consequence of the asserted minimum thirteen.

Reproduction and trust. Run `lean Main.lean` with Lean 4.19.0. Finite checks use kernel-checked `decide`; no `native_decide`, unproved placeholder, or additional axiom is used. The file prints axiom dependencies for its main nonexistence, criticality, spectrum, and nonisomorphism results.